

CS 466: Homework 6 — Solutions

Problem 1

Part A (40 points) — IND-CPA attack

Adversary. Fix any two distinct 4-block plaintexts $m_0, m_1 \in \{0, 1\}^{512}$ (for concreteness, $m_0 = 0^{512}$ and $m_1 = 0^{384} \| 1 \| 0^{127}$, which differ only in the first bit of the last block).

Adversary $A^{\text{LR}(\cdot, \cdot)}$:

$(C_e^{(1)}, T^{(1)}) \leftarrow \text{LR}(m_0, m_0)$

$(C_e^{(2)}, T^{(2)}) \leftarrow \text{LR}(m_0, m_1)$

If $T^{(1)} = T^{(2)}$ then Ret 0

Else Ret 1

Claim: $\text{Adv}_{\text{SE}}^{\text{ind-cpa}}(A) = \Pr[\text{Right}_{\text{SE}}^A \Rightarrow 1] - \Pr[\text{Left}_{\text{SE}}^A \Rightarrow 1] = 1$.

Proof. The tag depends only on the plaintext (not on the random IV $C_e[0]$), and m_0, m_1 agree on blocks 1, 2, 3 and differ only in block 4.

Right game. Query 1 encrypts m_0 and query 2 encrypts m_1 . Let Y_i (resp. Y'_i) denote the CBC-MAC chaining values on m_0 (resp. m_1). Since m_0 and m_1 agree on blocks 1, 2, 3, we have $Y_1 = Y'_1$, $Y_2 = Y'_2$, $Y_3 = Y'_3$, so $T^{(1)} = \text{AES}_K(Y_3 \oplus m_0[4])$ and $T^{(2)} = \text{AES}_K(Y_3 \oplus m_1[4])$. Since $m_0[4] \neq m_1[4]$ and AES_K is a permutation, $T^{(1)} \neq T^{(2)}$, so A returns 1.

Left game. Both queries encrypt m_0 , so $T^{(1)} = T^{(2)} = \text{CBC-MAC}_K(m_0)$ and A returns 0.

The advantage claim follows: $\text{Adv}_{\text{SE}}^{\text{ind-cpa}}(A) = 1 - 0 = 1$.

Resources: 2 LR queries, $O(1)$ work.

Part B (40 points) — INT-CTXT attack

Adversary. Consider the adversary A defined as follows:

Adversary $A^{\text{ENC}(\cdot)}$:

$C_e^* \leftarrow 0^{640}; \quad T^* \leftarrow 0^{128}$

Ret (C_e^*, T^*)

Claim: $\text{Adv}_{\text{SE}}^{\text{int-ctxt}}(A) = 1$, namely $\Pr[\mathcal{D}_K(C_e^*, T^*) \neq \perp] = 1$ and $\Pr[C_e^* \notin \mathcal{S}] = 1$.

Proof. Parse $C_e^* = C_e^*[0] \| C_e^*[1] \| \dots \| C_e^*[4]$ with each $C_e^*[i] = 0^{128}$, and let $P = \text{AES}_K^{-1}(0^{128})$. The decryption algorithm computes, for $i = 1, 2, 3, 4$,

$$M^*[i] = \text{AES}_K^{-1}(C_e^*[i]) \oplus C_e^*[i-1] = \text{AES}_K^{-1}(0^{128}) \oplus 0^{128} = P,$$

so $M^* = P \| P \| P \| P$. Then $C_m[1] = \text{AES}_K(0^{128} \oplus P) = \text{AES}_K(P) = 0^{128}$, and by iterating $C_m[i] = \text{AES}_K(0^{128} \oplus P) = 0^{128}$ for every $i \in \{1, 2, 3, 4\}$. Hence $C_m[4] = 0^{128} = T^*$, so verification succeeds and $\mathcal{D}_K(C_e^*, T^*) \neq \perp$ regardless of K .

A makes no ENC queries, so $\mathcal{S} = \emptyset$ and $C_e^* \notin \mathcal{S}$, so the advantage claim follows.

Resources: $O(1)$ work.

Part C (20 points) — Is SE an Encrypt-and-MAC scheme?

Answer. *No.* Recall that any generic composition method takes two ingredients, a symmetric encryption scheme SE_0 with key space $\mathcal{K}_e$ and message authentication code with key space $\mathcal{K}_m$, to produce a symmetric encryption scheme $\text{AE} = (\mathcal{K}^*, \mathcal{E}^*, \mathcal{D}^*)$ such that

$$\mathcal{K}^* : K_e \leftarrow \$ \mathcal{K}_e; K_m \leftarrow \$ \mathcal{K}_m; \text{ return } (K_e, K_m).$$

In particular, the two underlying keys are generated *independently* by the key-generation algorithm of the composed scheme. In SE, however, the key-generation algorithm samples a *single* 128-bit key K and *reuses* this key for both CBC\$-encryption and basic CBC-MAC.

Bonus Problem

Consider the modified scheme $\text{SE}' = (\mathcal{K}', \mathcal{E}', \mathcal{D}')$ which is identical to SE except that $\mathcal{K}'$ samples two *independent* keys $K_e, K_m \leftarrow \$ \{0, 1\}^{128}$.

B1 — Is SE' IND-CPA secure?

Answer: *No.* The solution is essentially the same as in Problem 1 part A above.

B2 — Is SE' INT-CTXT secure?

Answer: *No.*^a

Idea: When some 4-block message M is queried, the encryption procedure returns $C_e = \text{IV} \| Z_1 \| Z_2 \| Z_3 \| Z_4$ where $Z_i = \text{AES}_{K_e}(Z_{i-1} \oplus M[i])$. Note that $\text{IV}' \| Z_1 \| Z_2 \| Z_3 \| Z_4$ is also a valid CBC ciphertext, except the decrypted first plaintext block is shifted by $\text{IV} \oplus \text{IV}'$. So our adversary swaps the IV to 0^{128} to produce a fresh ciphertext that decrypts to a different plaintext, then queries the encryption oracle for the tag of that different plaintext.

Concretely, take $M = 0^{512}$. Then $Z_i = \text{AES}_{K_e}^i(\text{IV}_1)$ and $0^{128} \| Z_1 \| \dots \| Z_4$ decrypts to $\text{IV}_1 \| 0^{384}$.

Adversary:

Adversary $A^{\text{ENC}(\cdot)}$:

$(C_e^{(1)}, T^{(1)}) \leftarrow \text{ENC}(0^{512})$
 Parse $C_e^{(1)}$ as $\text{IV}_1 \| Z_1 \| Z_2 \| Z_3 \| Z_4$
 $(C_e^{(2)}, T^{(2)}) \leftarrow \text{ENC}(\text{IV}_1 \| 0^{384})$
 $C_e^* \leftarrow 0^{128} \| Z_1 \| Z_2 \| Z_3 \| Z_4$
 Ret $(C_e^*, T^{(2)})$

^aAttack due to Andrew Friedman.

Claim: $\text{Adv}_{\text{SE}'}^{\text{int-ctxt}}(A) \geq (1 - 2^{-128})^2 \geq 1 - 2^{-127}$.

Claim: $\Pr[\mathcal{D}'(C_e^*, T^{(2)}) \neq \perp] = 1$ and $\Pr[C_e^* \notin \mathcal{S}] \geq (1 - 2^{-128})^2$.

Proof. Decrypting under K_e with $C_e^*[0] = 0^{128}$ and $C_e^*[i] = Z_i$ for $i \in \{1, 2, 3, 4\}$ gives $M^*[1] = \text{AES}_{K_e}^{-1}(Z_1) \oplus 0^{128} = \text{IV}_1$ (using $Z_1 = \text{AES}_{K_e}(\text{IV}_1)$ from query 1) and $M^*[i] = \text{AES}_{K_e}^{-1}(Z_i) \oplus Z_{i-1} = Z_{i-1} \oplus Z_{i-1} = 0^{128}$ for $i = 2, 3, 4$. So $M^* = \text{IV}_1 \| 0^{384}$, which is exactly the plaintext of query 2; the tag check evaluates $\text{CBC-MAC}_{K_m}(M^*) = T^{(2)}$, so verification passes.

Note $\mathcal{S} = \{C_e^{(1)}, C_e^{(2)}\}$. We consider each ciphertext and show it is not equal to C_e^* . In the first case, $C_e^* = C_e^{(1)} \iff \text{IV}_1 = 0^{128}$, which occurs with probability 2^{-128} . In the second case, $C_e^* = C_e^{(2)}$ implies $Z_1 = C_e^{(2)}[1] = \text{AES}_{K_e}(\text{IV}_2 \oplus \text{IV}_1)$. Since $Z_1 = \text{AES}_{K_e}(\text{IV}_1)$, this implies $\text{IV}_2 = 0^{128}$, which also occurs with probability 2^{-128} . So $\Pr[C_e^* \notin \mathcal{S}] \geq (1 - 2^{-128})^2$.

The advantage claim follows: $\text{Adv}_{\text{SE}'}^{\text{int-ctxt}}(A) = \Pr[\mathcal{D}'(C_e^*, T^{(2)}) \neq \perp \wedge C_e^* \notin \mathcal{S}] \geq 1 \cdot (1 - 2^{-128})^2$.

Resources: 2 ENC queries, $O(1)$ work.

Remark: This attack is a concrete instance of Bellare–Namprempre’s Theorem 4.4 in [1].

References

- [1] M. Bellare and C. Namprempre. Authenticated encryption: Relations among notions and analysis of the generic composition paradigm. In T. Okamoto, editor, *Advances in Cryptology — ASIACRYPT 2000*, volume 1976 of *LNCS*, pages 531–545. Springer, 2000.